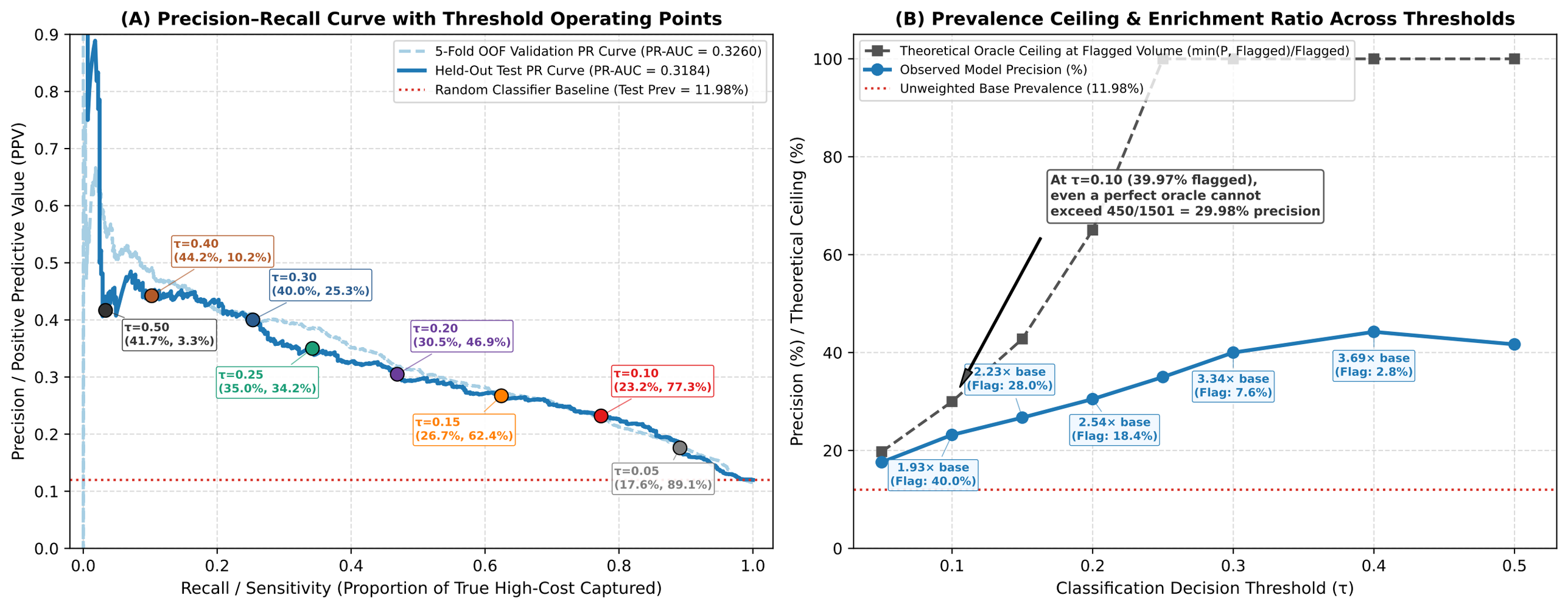

(A) Precision–Recall Curve with Threshold Operating Points
5-Fold OOF Validation PR Curve (PR-AUC = 0.3260)
Held-Out Test PR Curve (PR-AUC = 0.3184)
Random Classifier Baseline (Test Prev = 11.98%)
τ=0.40 (44.2%, 10.2%)
τ=0.30 (40.0%, 25.3%)
τ=0.50 (41.7%, 3.3%)
τ=0.25 (35.0%, 34.2%)
τ=0.20 (30.5%, 46.9%)
τ=0.10 (23.2%, 77.3%)
τ=0.15 (26.7%, 62.4%)
τ=0.05 (17.6%, 89.1%)
Precision / Positive Predictive Value (PPV)
Recall / Sensitivity (Proportion of True High-Cost Captured)
(B) Prevalence Ceiling & Enrichment Ratio Across Thresholds
Theoretical Oracle Ceiling at Flagged Volume (min(P, Flagged)/Flagged)
Observed Model Precision (%)
Unweighted Base Prevalence (11.98%)
At τ=0.10 (39.97% flagged), even a perfect oracle cannot exceed 450/1501 = 29.98% precision
2.23× base (Flag: 28.0%)
1.93× base (Flag: 40.0%)
2.54× base (Flag: 18.4%)
3.34× base (Flag: 7.6%)
3.69× base (Flag: 2.8%)
Precision (%) / Theoretical Ceiling (%)
Classification Decision Threshold (τ)